

Yudi (Mike) Qin

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EDUCATION

University of Illinois at Urbana-Champaign

Aug. 2022 - Dec. 2024

M.S. in Computer Science | Advisor: Nancy M. Amato

University of Illinois at Urbana-Champaign

Aug. 2018 - May 2022

B.S. in Computer Science + Geography & GIS | GPA: 3.95/4.0

RESEARCH INTERESTS

Task and Motion Planning, Trajectory Optimizations, Multi-Robot Systems, Learning for Planning

PUBLICATIONS

Peer Reviewed Journal

- **M. Qin**, I. Solis, J. D. Motes, M. Morales and N. M. Amato, “K-ARC: Adaptive Robot Coordination for Multi-Robot Kinodynamic Planning”, in IEEE Robotics and Automation Letters, vol. 10, no. 10, pp. 10562-10569, Oct. 2025.
- I solis, J, Motes, **M. Qin**, M. Morales and N. M. Amato, “Experience-based Subproblem Planning for Multi-Robot Motion Planning”, 2025. In Submission.
- I Solis, J. Motes, **M. Qin**, M. Morales and N. M. Amato, “Adaptive Robot Coordination: A Subproblem-Based Approach for Hybrid Multi-Robot Motion Planning”, in IEEE Robotics and Automation Letters, vol. 9, no. 8, pp. 7238-7245, Aug. 2024.

Thesis and Others

- **M. Qin**, I. Solis, J. D. Motes, M. Morales and N. M. Amato, “A Hybrid Framework for Multi-Robot Kinodynamic Planning”. Master’s Thesis. Champaign, IL. Dec. 2024.

EXPERIENCE

Ubtech Robotics Inc.

Feb. 2025 - Present

Robotics Algorithm Engineer, Autonomous Navigation (Full-Time)

Beijing, China

- Designed and implemented an autonomous parking module within the existing autonomous navigation stack while regularly maintaining the codebase.
- Implemented planning algorithms from state-of-the-art research papers in the free-space planning module, focusing on improving the overall accuracy and success rate in autonomous parking scenarios.
- Developed an Isaac Sim simulation framework alongside the existing Webots simulation framework, which supports data collections and algorithm testing for various modules in perception, planning, and control.

Parasol Lab

Aug. 2022 - Dec. 2024

Graduate Research Assistant

Champaign, IL

- Contributed to the development of the multi-robot system components for PPL(Parasol Planning Library), an open-source C++ robotics task and motion planning library
- Developed an algorithm for solving multi-robot motion planning problems, achieving an overall 20% speed-up over existing solutions across various test scenarios.
- Extending the algorithm to incorporate trajectory optimizations for real-time controls of the robots.

Foxconn Interconnect Technology

May 2023 - Aug. 2023

Software Development Engineering Intern

Remote

- Developed a software package in ROS for motion planning for multiple AGV(Automated Guided Vehicle) robots.
- Worked closely with my mentor to put this package into real usage for the company’s factory automation.
- Recognized as “Excellent Intern” for consistent contributions and performance during the internship.

Parasol Lab

May 2021 - Aug. 2021

Research Experience for Undergraduate (REU) Student

Champaign, IL

- Assisted in developing and implementing multi-robot motion planning algorithms in **C++**.
- Contributed to developing and open-sourcing PPL. Specifically, added testing suites for existing codes, refactored existing codes, and added new components to the library.

CyberGIS Center

Jun. 2020 - Mar. 2022

Student Intern

Champaign, IL

- Developed Python codes for processing large-scale geo-located data, gathered using Twitter's data collection API.
- Contributed to developing a web-based GIS application that is able to visualize and predict spatiotemporal patterns of infectious diseases.
- Implemented a spatial analytic tool known as MGWR (multi-scale geographically weighted regression) in Python and explored its application on disease data analysis.

TEACHING EXPERIENCE

University of Illinois at Urbana-Champaign

Aug. 2022 - Dec. 2024

Graduate Teaching Assistant

Champaign, IL

- CS 441, Applied Machine Learning (Fall 2022)
- CS 421, Programming Languages and Compilers (Spring and Summer 2023)
- CS 440, Intro to Artificial Intelligence (Spring 2024)

HONORS AND AWARDS

UIUC College of Liberal Arts and Science, Dean's List

Fall 2018 - Spring 2022

The Howard and Ruth Roepke Undergraduate Scholarship

Spring 2019